

Bayesian Curve Fitting by Gaussian Processes

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1 Brownian Motion and Integrated Brownian Motion

We studied the following model in the lectures. We have a response variable y and a single covariate x . In our example, y denotes weekly earnings and x denotes years of experience. The covariate x takes the values $0, 1, \dots, m$ for some fixed integer m .

Our data is $(x_i, y_i), i = 1, \dots, n$. We consider the model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 \text{ReLU}(x_i - 1) + \dots + \beta_m \text{ReLU}(x_i - (m - 1)) + \epsilon_i \quad (1)$$

where $\text{ReLU}(u) := \max(u, 0)$, and $\epsilon_i \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2)$.

We discussed Bayesian inference in this model using the prior:

$$\beta_0 \sim N(0, C), \beta_1 \sim N(0, C), \beta_2, \dots, \beta_m \stackrel{\text{i.i.d}}{\sim} N(0, \tau^2) \quad (2)$$

We remarked that the resulting prior on the regression function:

$$f(x) = \beta_0 + \beta_1 x + \beta_2 \text{ReLU}(x - 1) + \dots + \beta_m \text{ReLU}(x - (m - 1))$$

with prior (2) is very similar to:

$$f(x) = \beta_0 + \beta_1 x + \tau I(x) \quad (3)$$

where $I(x)$ is integrated Brownian motion (and $\beta_0, \beta_1 \stackrel{\text{i.i.d}}{\sim} N(0, C)$). We shall now see why the Integrated Brownian Motion is appearing here.

1.1 Defining Brownian Motion

Before defining Brownian motion, let us recall the definition of Brownian motion. Brownian motion on $[0, M]$ (for some fixed $M > 0$) is a stochastic process $\{B(t), 0 \leq t \leq M\}$ characterized by the following two properties:

1. Every realization is a continuous function on $[0, M]$, and $B(0) = 0$.
2. For every fixed $t_1, \dots, t_k \in [0, M]$, the random vector $(B(t_1), \dots, B(t_k))$ has a multivariate normal distribution with mean zero and covariance matrix

$$\text{Cov}(B(t_i), B(t_j)) = \min(t_i, t_j).$$

This definition suggests the following method for simulating $B(t)$. Construct the dense grid

$$0 = t_0 < t_1 < \dots < t_N = M$$

on $[0, M]$ with $t_i = iM/N$. One can then simulate $(B(t_1), \dots, B(t_N))$ from the multivariate normal distribution with mean zero and covariance matrix $\min(t_i, t_j)$.

However, this approach is computationally inefficient for large N . Sampling directly from an $N \times N$ covariance matrix typically requires a matrix factorization (such as a Cholesky decomposition), which has computational cost on the order of $O(N^3)$.

For simulations in particular, the following alternative definition is more appealing:

1. Every realization is a continuous function on $[0, M]$ with $B(0) = 0$.
2. The process has independent increments which means that $B(t_1), B(t_2) - B(t_1), \dots, B(t_k) - B(t_{k-1})$ are independent whenever $0 \leq t_1 < \dots < t_k \leq M$.
3. $B(t) - B(s) \sim N(0, t - s)$ whenever $0 \leq s \leq t$.

With this definition, we can simulate $B(t_1), \dots, B(t_N)$ for $t_i = iM/N$ in the following way:

- First generate $\alpha_1, \dots, \alpha_N \stackrel{\text{i.i.d}}{\sim} N(0, M/N)$
- Take $B(t_i) = \alpha_1 + \dots + \alpha_i$ for $i = 1, \dots, N$.

Note that in this way $\alpha_i = B(t_i) - B(t_{i-1})$. The equation $B(t_i) = \alpha_1 + \dots + \alpha_i$ can also be written as

$$B(t) = \sum_{i=1}^N \alpha_i I\{t_i \leq t\} \quad \text{for } t \in \{t_1, \dots, t_N\}.$$

Because of the denseness of the grid (when N is large), we are claiming that

$$B(t) \approx \sum_{i=1}^N \alpha_i I\{t_i \leq t\} \quad \text{for all } t \in [0, M]. \quad (4)$$

This means that Brownian motion can be well-approximated (when N is large) by a piecewise constant process which jumps at each grid point $t_i = iM/N$ by a small amount given by $N(0, M/N)$.

We are now ready to define the Integrated Brownian Motion. This is given by:

$$I(t) = \int_0^t B(s) ds. \quad (5)$$

Using the approximation (4), we can write

$$I(t) \approx \int_0^t \left(\sum_{i=1}^N \alpha_i I\{t_i \leq s\} \right) ds = \sum_{i=1}^N \alpha_i \int_0^t I\{t_i \leq s\} ds = \sum_{i=1}^N \alpha_i (t - t_i)_+.$$

Therefore Integrated Brownian Motion can be approximated as:

$$I(t) \approx \sum_{i=1}^N \alpha_i (t - t_i)_+ \quad (6)$$

when N is large, $t_i = iM/N$ and $\alpha_i \stackrel{\text{i.i.d.}}{\sim} N(0, M/N)$.

Using (6), the (prior) model (3) can therefore be written as:

$$f(x) \approx \beta_0 + \beta_1 x + \tau \sum_{i=1}^N \alpha_i \left(x - \frac{iM}{N} \right)_+$$

Here N is a large number. If we do a further coarse approximation with $N = M$, we get back the model (1) if we make the identification $\beta_i = \tau \alpha_{i-1}$ for $i = 2, \dots, N$ and $M = m$. Therefore, the model (1) can be understood as a coarse discretization of the model (3) which stipulates that f is an Integrated Brownian Motion (scaled by τ) and added to a linear function $\beta_0 + \beta_1 x$ with the uninformative $N(0, C)$ prior on the coefficients β_0, β_1 .

The prior (3) is an example of a Gaussian process prior.

2 Gaussian Process Regression

Consider the usual nonparametric regression problem where the goal is to estimate an unknown function $f : \Omega \rightarrow \mathbb{R}$ from observations $(x_1, y_1), \dots, (x_n, y_n)$ under the model:

$$y_i = f(x_i) + \epsilon_i \quad \text{where } \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

Here Ω denotes the domain which is a subset of $\mathbb{R}^d$ for some $d \geq 1$.

We assume that $\{f(x), x \in \Omega\}$ forms a Gaussian process with mean function $\mu(x), x \in \Omega$ and covariance function or *kernel* given by $K(x, x')$ i.e.,

$$\text{Cov}(f(x), f(x')) = K(x, x') \quad \text{for all } x, x' \in \Omega.$$

The kernel is positive semi-definite i.e., for every $N \geq 1$, distinct points $u_1, \dots, u_N \in \Omega$, the $N \times N$ matrix with $(i, j)^{th}$ entry $K(u_i, u_j)$ is positive semi-definite. Often, this matrix will be positive definite, and hence invertible.

The mean function is usually taken to be zero. Here are some standard special cases. Here are some examples of Gaussian processes and kernels:

1. **Brownian Motion:** Here $\Omega = [0, \infty)$ and $K(s, t) = \min(s, t)$.
2. **(scaled) Brownian Motion plus constant:** Here $\Omega = [0, \infty)$ and we assume that $f(t) = \beta_0 + \tau W_t$ where $W_t \sim BM$, $\beta_0 \sim N(0, C)$ and $\tau > 0$ (and independence between β_0 and $\{W_t\}$). C will be taken to be large. Now

$$K(s, t) = C + \tau^2 \min(s, t).$$

3. **Integrated Brownian Motion:** $\Omega = [0, \infty)$ and

$$f(t) = \int_0^t W_s ds \quad \text{where } W_s \sim BM.$$

The kernel is given by

$$\begin{aligned} K(s, t) &= \text{Cov} \left(\int_0^s W_u du, \int_0^t W_v dv \right) \\ &= \int_0^s \int_0^t \text{Cov}(W_u, W_v) dv du = \int_0^s \int_0^t \min(u, v) dv du \end{aligned}$$

To simplify further, assume that $s \leq t$ so that

$$\begin{aligned} K(s, t) &= \int_0^s \left(\int_0^u v dv + \int_u^t u dv \right) du \\ &= \int_0^s \left(\frac{u^2}{2} + u(t - u) \right) du = \int_0^s \left(ut - \frac{u^2}{2} \right) du = t \frac{s^2}{2} - \frac{s^3}{6}. \end{aligned}$$

For general s and t , we have

$$K(s, t) = \frac{1}{2} \max(s, t) (\min(s, t))^2 - \frac{1}{6} (\min(s, t))^3.$$

4. **(scaled) Integrated Brownian Motion Plus a Linear Term:** In the IBM model, we have $f(0) = 0$ and $f'(0) = 0$. This might be an unrealistic assumption to make when f is completely unknown. In this case, a better model might be

$$f(t) = \beta_0 + \beta_1 t + \tau \int_0^t W_s ds$$

where $\beta_0, \beta_1, \{W_s\}$ are independent with W_s being Brownian motion and $\beta_0, \beta_1 \stackrel{\text{i.i.d}}{\sim} N(0, C)$. The kernel now becomes

$$\begin{aligned} K(s, t) &= \text{Cov} \left(\beta_0 + \beta_1 s + \tau \int_0^s W_u du, \beta_0 + \beta_1 t + \tau \int_0^t W_v dv \right) \\ &= C(1 + st) + \frac{\tau^2}{2} \max(s, t) (\min(s, t))^2 - \frac{\tau^2}{6} (\min(s, t))^3. \end{aligned}$$

Show samples from BM and IBM with random linear drift.

There are many other examples of kernels. Next we see some basic applications of Gaussian Processes.

3 Bayesian Numerical Analysis

3.1 Interpolation

Suppose we are given values of f at $0 \leq x_1 < \dots < x_n \leq 1$: $f(x_1), \dots, f(x_n)$. Let $x \in [0, 1]$ be a new point (i.e., distinct from $x_1, \dots, x_n$). What can we say about $f(x)$?

We will solve this problem assuming a Gaussian process prior for f with mean zero and covariance kernel $K(\cdot, \cdot)$. The answer then will be given by the conditional distribution of $f(x)$ given $f(x_1), \dots, f(x_n)$. To compute this, we note that:

$$(f(x_1), \dots, f(x_n), f(x))^T \sim N \left(0, \begin{pmatrix} (K(x_i, x_j))_{n \times n} & (K(x_i, x))_{n \times 1} \\ (K(x, x_i))_{1 \times n} & K(x, x) \end{pmatrix} \right).$$

Using the notation $K = (K(x_i, x_j))_{n \times n}$ and $\mathbf{k} = (K(x_i, x))_{n \times 1}$, we can write the conditional distribution of $f(x)$ given $f(x_1), \dots, f(x_n)$ as

$$f(x) \mid f(x_1), \dots, f(x_n) \sim N(\mathbf{k}^T K^{-1}(f(x_1), \dots, f(x_n))^T, K(x, x) - \mathbf{k}^T K^{-1} \mathbf{k}).$$

Thus the posterior mean (or mode) estimate of $f(x)$ is given by

$$\widehat{f(x)} = \mathbf{k}^T K^{-1}(f(x_1), \dots, f(x_n))^T. \quad (7)$$

For different kernels K , the above expression behaves differently. The simplest example is that of Brownian Motion.

Example 3.1 (Brownian Motion). Suppose $f(x) = \beta_0 + \tau B(x)$ where $B(x)$ is Brownian Motion, $\beta_0 \sim N(0, C)$ with $C \rightarrow \infty$ and τ is a fixed positive constant. Then the kernel is:

$$K(x_1, x_2) = C + \tau^2 \min(x_1, x_2). \quad (8)$$

In this case, (7) can be explicitly evaluated (in the limit $C \rightarrow \infty$) as

$$\widehat{f(x)} = \begin{cases} f(x_1), & x \leq x_1, \\ \frac{x_{i+1} - x}{x_{i+1} - x_i} f(x_i) + \frac{x - x_i}{x_{i+1} - x_i} f(x_{i+1}), & x_i \leq x \leq x_{i+1}, \\ f(x_n), & x \geq x_n. \end{cases} \quad (9)$$

Exercise: Can you show why (7) with the kernel (8) leads to (9)?

The formula (9) for $\widehat{f(x)}$ performs linear interpolation between the observed points, and constant extrapolation outside the observed range. Also note that $\widehat{f(x)}$ in (9) does not depend on τ . The hyperparameter τ only affects the posterior variance of $f(x)$, and not the posterior mean.

Next example is that of the Integrated Brownian Motion prior.

Example 3.2 (Integrated Brownian Motion). Suppose

$$f(x) = \beta_0 + \beta_1 x + \tau I(x), \quad (10)$$

where $I(x)$ is integrated Brownian motion and $\beta_0, \beta_1 \stackrel{i.i.d.}{\sim} N(0, C)$ with $C \rightarrow \infty$. In this case, the posterior mean in (7) can be evaluated explicitly and is the natural cubic spline interpolant of the observations. We will discuss more about splines later.

3.2 Integration

Suppose we are given values of f at $0 \leq x_1 < \dots < x_n \leq 1$: $f(x_1), f(x_2), \dots, f(x_n)$. What can we say about $\int_0^1 f(x)dx$?

Again we place a Gaussian process prior on f . By linearity, we can write

$$\begin{aligned} & \left[\int_0^1 f(x)dx \mid f(x_1), \dots, f(x_n) \right] \\ &= \int_0^1 [f(x) \mid f(x_1), \dots, f(x_n)] \\ &= \int_0^1 \mathbf{k}^T K^{-1}(f(x_1), \dots, f(x_n))^T \\ &= \left(\int_0^1 k(x_1, x)dx, \dots, \int_0^1 k(x_n, x)dx \right)^T K^{-1}(f(x_1), \dots, f(x_n))^T. \end{aligned}$$

Example 3.3 (Brownian Motion). For the Brownian motion, we already have the formula (9) for $[f(x) \mid f(x_1), \dots, f(x_n)]$. So the posterior mean for $\int_0^1 f(x)dx$ is obtained by simply integrating the right hand side of (9) from 0 to 1. This leads to:

$$\begin{aligned} & \left(\int_0^1 f(x)dx \mid f(x_1), \dots, f(x_n) \right) \\ &= x_1 f(x_1) + \sum_{i=1}^{n-1} (x_{i+1} - x_i) \frac{f(x_i) + f(x_{i+1})}{2} + (1 - x_n) f(x_n). \end{aligned}$$

This is the trapezoidal integration rule. Therefore, with the Brownian motion prior, the posterior mean estimate of $\int_0^1 f(x)dx$ coincides with the trapezoidal rule.

Example 3.4 (Integrated Brownian Motion). With the prior (10), we get a version of the Simpson integration rule.

For more on the relation between classical integration (quadrature) rules and Gaussian processes, see the book [Hennig et al. \[2022\]](#) or the paper [Diaconis \[1988\]](#).

In the next section, we shall see how to use Gaussian processes for the regression problem.

Show plots for interpolation with BM and IBM with constant and linear drifts.

4 Nonparametric Regression with Gaussian Processes

We will see how to perform regression using Gaussian processes. The key difference between **regression** and **interpolation** is that, in regression, the values $f(x_1), \dots, f(x_n)$ are not observed exactly; instead, they are observed with noise.

More precisely, we are given observations $y_1, \dots, y_n$ modeled as

$$y_i = f(x_i) + \epsilon_i, \quad \text{where } \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

The parameter σ controls the level of noise. As in the interpolation setting, our goal is to estimate $f(x)$ at a test point x . This test point may be different from the observed inputs $x_1, \dots, x_n$, or it may coincide with one of them.

Importantly, because the observations are noisy, it is meaningful to estimate $f(x_i)$ even at observed points. In fact, by combining information from all observations, we can often obtain a

better estimate of $f(x_i)$ than the raw observation y_i . Another key difference from interpolation is that the input points $x_1, \dots, x_n$ need not necessarily be distinct. Since each observation contains noise, having repeated measurements at the same input can help improve the overall estimate.

The solution to the regression problem is very similar to that of the interpolation problem with only one difference. The goal is to estimate $f(x)$ at the test point x given the data $(x_1, y_1), \dots, (x_n, y_n)$. We will use the conditional distribution of $f(x)$ given $y_1, \dots, y_n$ (we will assume that $x_1, \dots, x_n, x$ are deterministic). To calculate this conditional distribution, first note that the marginal distribution of $(y_1, \dots, y_n, f(x))$ is given by

$$(y_1, \dots, y_n, f(x)) \sim N \left(0, \begin{pmatrix} (K(x_i, x_j))_{n \times n} + \sigma^2 I_n & (K(x_i, x))_{n \times 1} \\ (K(x, x_i))_{1 \times n} & K(x, x) \end{pmatrix} \right).$$

Using the notation $K = (K(x_i, x_j))_{n \times n}$ and $\mathbf{k} = (K(x_i, x))_{n \times 1}$, we can write the conditional distribution of $f(x)$ given $y_1, \dots, y_n$ as

$$f(x) \mid \text{data} \sim N \left(\mathbf{k}^T (K + \sigma^2 I_n)^{-1} y, K(x, x) - \mathbf{k}^T (K + \sigma^2 I_n)^{-1} \mathbf{k} \right).$$

Thus the posterior mean (or mode) estimate of $f(x)$ is given by

$$\widehat{f(x)} = \mathbf{k}^T (K + \sigma^2 I_n)^{-1} y, \quad (11)$$

where y is the $n \times 1$ vector with entries $y_1, \dots, y_n$.

So the only difference between (11) and (7) is the presence of the additional $\sigma^2 I_n$ term in case of regression.

Often, we would need to estimate σ from the observed data (and also additional hyperparameters present in the kernel K). For these, the marginal likelihood of $y_1, \dots, y_n$ is important. This is simply the multivariate normal distribution with mean vector 0 and covariance $K + \sigma^2 I_n$. To illustrate the calculations, let us take the special case of the Integrated Brownian Motion prior.

5 Gaussian Process Regression with the Integrated Brownian Motion Prior

We now specialize the general Gaussian process regression framework to the integrated Brownian motion (IBM) prior introduced in the previous section. Besides being one of the simplest nontrivial Gaussian processes, the IBM prior is closely connected with ReLU basis functions and with natural cubic smoothing splines, as we have seen.

Recall that our prior is

$$f(x) = \beta_0 + \beta_1 x + \tau I(x), \quad (12)$$

where $I(x)$ denotes integrated Brownian motion, β_0, β_1 are independent $N(0, C)$ random variables, and C is intended to be very large (ideally $C = \infty$). The parameter $\tau > 0$ controls the size of the stochastic component of the prior.

Suppose that we observe

$$(x_1, y_1), \dots, (x_n, y_n)$$

from the regression model

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

Our goal is to compute the posterior distribution of $f(x)$ at a new input x . By the general Gaussian process theory developed in the previous section, the posterior mean is

$$\hat{f}(x) = \mathbf{k}^T (K + \sigma^2 I_n)^{-1} y, \quad (13)$$

where

- K is the covariance matrix of $(f(x_1), \dots, f(x_n))$,
- $\mathbf{k}$ is the covariance vector between $f(x)$ and $(f(x_1), \dots, f(x_n))$,
- and $y = (y_1, \dots, y_n)^T$.

Thus everything reduces to calculating the covariance kernel corresponding to the prior (12). The covariance kernel of integrated Brownian motion is

$$K_I(u, v) = \frac{1}{2}(\min(u, v))^2 \max(u, v) - \frac{1}{6}(\min(u, v))^3.$$

Since the linear drift and the integrated Brownian motion are independent, the covariance kernel of f is

$$K(u, v) = C(1 + uv) + \tau^2 K_I(u, v). \quad (14)$$

To simplify the notation, let X denote the usual design matrix for simple linear regression,

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix},$$

and let K_I also denote the $n \times n$ matrix with (i, j) -entry $K_I(x_i, x_j)$.

Using (14), the covariance matrix of $(f(x_1), \dots, f(x_n))$ can be written compactly as

$$K = CXX^T + \tau^2 K_I. \quad (15)$$

Similarly, if

$$\mathbf{k}_I = (K_I(x, x_1), \dots, K_I(x, x_n))^T,$$

then

$$\mathbf{k} = CX(1, x)^T + \tau^2 \mathbf{k}_I. \quad (16)$$

Substituting (15) and (16) into (13) gives

$$\hat{f}(x) = (C(1, x)X^T + \tau^2 \mathbf{k}_I^T) (CXX^T + \tau^2 K_I + \sigma^2 I_n)^{-1} y. \quad (17)$$

Equation (17) is already an explicit expression for the posterior mean. However, it still depends on the large constant C . Since C is intended to represent a diffuse prior on the intercept and slope, we will try to take the limit of this expression as $C \rightarrow \infty$. To do this, we first apply the Sherman–Morrison–Woodbury identity to simplify the matrix inverse and then take the limit.

Define

$$\Sigma = \tau^2 K_I + \sigma^2 I_n.$$

Then

$$CXX^T + \tau^2 K_I + \sigma^2 I_n = CXX^T + \Sigma.$$

Applying the Sherman–Morrison–Woodbury identity gives

$$(CXX^T + \Sigma)^{-1} = \Sigma^{-1} - \Sigma^{-1}X(C^{-1}I_2 + X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}. \quad (18)$$

Substituting (18) into (17), we obtain

$$\begin{aligned} \hat{f}(x) &= (C(1, x)X^T + \tau^2 \mathbf{k}_I^T) \\ &\quad \times \left[\Sigma^{-1} - \Sigma^{-1}X(C^{-1}I_2 + X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1} \right] y. \end{aligned} \quad (19)$$

We now let $C \rightarrow \infty$, corresponding to a diffuse prior on the intercept and slope. It is convenient to reparametrize the model in terms of

$$\gamma = \frac{\tau}{\sigma},$$

which measures the relative size of the IBM component compared with the observation noise. Since

$$\Sigma = \tau^2 K_I + \sigma^2 I_n = \sigma^2 (I_n + \gamma^2 K_I),$$

define

$$A_\gamma = I_n + \gamma^2 K_I.$$

The following theorem gives the limiting posterior mean.

Theorem 5.1. *As $C \rightarrow \infty$, the posterior mean (19) converges to*

$$\begin{aligned} \hat{f}(x) &= (1, x)(X^T A_\gamma^{-1} X)^{-1} X^T A_\gamma^{-1} y \\ &\quad + \gamma^2 \mathbf{k}_I^T (A_\gamma^{-1} - A_\gamma^{-1} X (X^T A_\gamma^{-1} X)^{-1} X^T A_\gamma^{-1}) y. \end{aligned} \quad (20)$$

Notice that the posterior mean depends on the hyperparameters only through the ratio $\frac{\tau}{\sigma}$ which therefore plays the role of the smoothing parameter in the model.

5.1 Estimation of the Hyperparameters

Up to this point, we have assumed that the hyperparameters τ and σ are known. In practice, however, these quantities have to be estimated from the observed data. It is more convenient to work with γ and σ with $\gamma = \frac{\tau}{\sigma}$ since the posterior mean derived in Theorem 5.1 depends only on γ .

There are two common approaches to estimating these hyperparameters. The first is the *empirical Bayes* approach, in which (γ, σ) are estimated by maximizing the marginal likelihood of the observed data. The second is the fully Bayesian approach, in which priors are placed on γ and σ , and inference is based on their posterior distribution. We derive the relevant quantities below.

Since

$$y = f + \varepsilon,$$

where

$$f \sim N(0, K), \quad \varepsilon \sim N(0, \sigma^2 I_n),$$

and f and ε are independent, we have

$$y \mid \tau, \sigma \sim N(0, K + \sigma^2 I_n).$$

Hence the marginal likelihood is

$$f(y \mid \tau, \sigma) \propto |K + \sigma^2 I_n|^{-1/2} \exp \left\{ -\frac{1}{2} y^T (K + \sigma^2 I_n)^{-1} y \right\}.$$

Step 1: Take limit as $C \rightarrow \infty$

To simplify this expression, recall that

$$K = CXX^T + \tau^2 K_I,$$

and define

$$\Sigma = \tau^2 K_I + \sigma^2 I_n.$$

Then

$$K + \sigma^2 I_n = \Sigma + CXX^T.$$

Using the Sherman–Morrison–Woodbury identity,

$$(K + \sigma^2 I_n)^{-1} = \Sigma^{-1} - \Sigma^{-1} X (C^{-1} I_2 + X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1}.$$

Letting $C \rightarrow \infty$ gives

$$(K + \sigma^2 I_n)^{-1} \longrightarrow \Sigma^{-1} - \Sigma^{-1} X (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1}. \quad (21)$$

Next, the matrix determinant lemma gives

$$\begin{aligned} |K + \sigma^2 I_n| &= |\Sigma + CXX^T| \\ &= |\Sigma| |I_2 + CX^T \Sigma^{-1} X|. \end{aligned}$$

As $C \rightarrow \infty$,

$$|I_2 + CX^T \Sigma^{-1} X| \sim C^2 |X^T \Sigma^{-1} X|,$$

where the factor C^2 is independent of (γ, σ) and may therefore be absorbed into the normalizing constant. Consequently,

$$|K + \sigma^2 I_n| \propto |\Sigma| |X^T \Sigma^{-1} X|. \quad (22)$$

Combining (21) and (22) yields the marginal likelihood

$$\begin{aligned} f(y \mid \tau, \sigma) &\propto |\Sigma|^{-1/2} |X^T \Sigma^{-1} X|^{-1/2} \\ &\times \exp \left[-\frac{1}{2} y^T \{ \Sigma^{-1} - \Sigma^{-1} X (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} \} y \right]. \end{aligned}$$

Step 2: Write the likelihood in terms of $\sigma, \gamma = \frac{\tau}{\sigma}$

Now write

$$\tau = \gamma \sigma.$$

Then

$$\Sigma = \sigma^2 (I_n + \gamma^2 K_I) = \sigma^2 A_\gamma,$$

where

$$A_\gamma = I_n + \gamma^2 K_I.$$

For convenience, define

$$Q_\gamma = y^T \left[A_\gamma^{-1} - A_\gamma^{-1} X (X^T A_\gamma^{-1} X)^{-1} X^T A_\gamma^{-1} \right] y. \quad (23)$$

Since

$$\Sigma = \sigma^2 A_\gamma, \quad (24)$$

we have

$$\Sigma^{-1} = \sigma^{-2} A_\gamma^{-1}. \quad (25)$$

Also, because Σ is an $n \times n$ matrix,

$$\begin{aligned} |\Sigma| &= |\sigma^2 A_\gamma| \\ &= (\sigma^2)^n |A_\gamma| \\ &= \sigma^{2n} |A_\gamma|. \end{aligned} \quad (26)$$

Therefore,

$$|\Sigma|^{-1/2} = \sigma^{-n} |A_\gamma|^{-1/2}. \quad (27)$$

Next,

$$\begin{aligned} X^T \Sigma^{-1} X &= X^T (\sigma^{-2} A_\gamma^{-1}) X \\ &= \sigma^{-2} X^T A_\gamma^{-1} X. \end{aligned} \quad (28)$$

Since X has two columns, corresponding to the intercept and linear term, the matrix $X^T \Sigma^{-1} X$ is 2×2 . Hence,

$$\begin{aligned} |X^T \Sigma^{-1} X| &= |\sigma^{-2} X^T A_\gamma^{-1} X| \\ &= \sigma^{-4} |X^T A_\gamma^{-1} X|. \end{aligned} \quad (29)$$

It follows that

$$|X^T \Sigma^{-1} X|^{-1/2} = \sigma^2 |X^T A_\gamma^{-1} X|^{-1/2}. \quad (30)$$

Combining the two determinant terms gives

$$\begin{aligned} |\Sigma|^{-1/2} |X^T \Sigma^{-1} X|^{-1/2} &= \sigma^{-n} |A_\gamma|^{-1/2} \sigma^2 |X^T A_\gamma^{-1} X|^{-1/2} \\ &= \sigma^{-(n-2)} |A_\gamma|^{-1/2} |X^T A_\gamma^{-1} X|^{-1/2}. \end{aligned} \quad (31)$$

We now simplify the quadratic form in the exponential. First,

$$\Sigma^{-1} = \sigma^{-2} A_\gamma^{-1}. \quad (32)$$

Moreover,

$$\begin{aligned} (X^T \Sigma^{-1} X)^{-1} &= (\sigma^{-2} X^T A_\gamma^{-1} X)^{-1} \\ &= \sigma^2 (X^T A_\gamma^{-1} X)^{-1}. \end{aligned} \quad (33)$$

Therefore,

$$\begin{aligned} \Sigma^{-1} X (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} &= (\sigma^{-2} A_\gamma^{-1}) X \left[\sigma^2 (X^T A_\gamma^{-1} X)^{-1} \right] X^T (\sigma^{-2} A_\gamma^{-1}) \\ &= \sigma^{-2} A_\gamma^{-1} X (X^T A_\gamma^{-1} X)^{-1} X^T A_\gamma^{-1}. \end{aligned} \quad (34)$$

Hence,

$$\begin{aligned}\Sigma^{-1} - \Sigma^{-1}X(X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1} \\ = \frac{1}{\sigma^2} \left[A_\gamma^{-1} - A_\gamma^{-1}X(X^TA_\gamma^{-1}X)^{-1}X^TA_\gamma^{-1} \right].\end{aligned}\quad (35)$$

It follows that

$$\begin{aligned}y^T \left\{ \Sigma^{-1} - \Sigma^{-1}X(X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1} \right\} y \\ = \frac{1}{\sigma^2} y^T \left[A_\gamma^{-1} - A_\gamma^{-1}X(X^TA_\gamma^{-1}X)^{-1}X^TA_\gamma^{-1} \right] y \\ = \frac{Q_\gamma}{\sigma^2}.\end{aligned}\quad (36)$$

The likelihood can then be written as

$$f(y \mid \gamma, \sigma) \propto \sigma^{-(n-2)} |A_\gamma|^{-1/2} |X^TA_\gamma^{-1}X|^{-1/2} \exp\left(-\frac{Q_\gamma}{2\sigma^2}\right). \quad (37)$$

Step 3: Put priors on σ, γ

We place the standard scale-invariant prior on σ ,

$$p(\sigma) \propto \frac{1}{\sigma}, \quad \sigma > 0, \quad (38)$$

and assign a prior density $p(\gamma)$ to γ . We assume that the priors on σ and γ are independent.

The joint density of y, σ, γ is therefore

$$\begin{aligned}p(y, \sigma, \gamma) &= f(y \mid \sigma, \gamma) p(\sigma) p(\gamma) \\ &\propto p(\gamma) \sigma^{-(n-1)} |A_\gamma|^{-1/2} |X^TA_\gamma^{-1}X|^{-1/2} \exp\left(-\frac{Q_\gamma}{2\sigma^2}\right).\end{aligned}\quad (39)$$

Step 4: From the joint density, observe the posterior distribution of σ is Inverse Gamma

Conditioning on y and γ , all factors that do not depend on σ can be absorbed into the conditional normalizing constant. Thus,

$$p(\sigma \mid y, \gamma) \propto \sigma^{-(n-1)} \exp\left(-\frac{Q_\gamma}{2\sigma^2}\right), \quad \sigma > 0. \quad (40)$$

To express this distribution as an inverse-Gamma distribution, define

$$v = \sigma^2. \quad (41)$$

Then

$$\sigma = v^{1/2}, \quad \left| \frac{d\sigma}{dv} \right| = \frac{1}{2} v^{-1/2}. \quad (42)$$

Using the change-of-variables formula,

$$\begin{aligned}p(v \mid y, \gamma) &= p\left(\sigma = v^{1/2} \mid y, \gamma\right) \left| \frac{d\sigma}{dv} \right| \\ &\propto \left(v^{1/2}\right)^{-(n-1)} \exp\left(-\frac{Q_\gamma}{2v}\right) v^{-1/2} \\ &= v^{-n/2} \exp\left(-\frac{Q_\gamma}{2v}\right).\end{aligned}\quad (43)$$

An inverse-Gamma distribution with shape parameter a and scale parameter b has density

$$f(v) \propto v^{-(a+1)} \exp\left(-\frac{b}{v}\right), \quad v > 0. \quad (44)$$

Comparing the two kernels gives

$$a = \frac{n}{2} - 1, \quad b = \frac{Q_\gamma}{2}. \quad (45)$$

Therefore,

$$\sigma^2 \mid y, \gamma \sim \text{Inv-Gamma}\left(\frac{n}{2} - 1, \frac{Q_\gamma}{2}\right), \quad (46)$$

where the second parameter is the scale parameter.

Equivalently,

$$\frac{1}{\sigma^2} \mid y, \gamma \sim \text{Gamma}\left(\frac{n}{2} - 1, \frac{Q_\gamma}{2}\right), \quad (47)$$

where the second parameter is the rate parameter.

Step 5: Integrate out σ from the joint density and obtain the posterior density of γ

To obtain the marginal posterior distribution of γ , we integrate σ out of the joint posterior. From (39), we have

$$\begin{aligned} p(\gamma \mid y) &\propto \int_0^\infty p(y, \sigma, \gamma) d\sigma \\ &\propto p(\gamma) |A_\gamma|^{-1/2} |X^T A_\gamma^{-1} X|^{-1/2} \\ &\quad \times \int_0^\infty \sigma^{-(n-1)} \exp\left(-\frac{Q_\gamma}{2\sigma^2}\right) d\sigma. \end{aligned} \quad (48)$$

To evaluate the integral, make the change of variables

$$u = \frac{1}{\sigma^2}. \quad (49)$$

Then

$$\sigma = u^{-1/2}, \quad d\sigma = -\frac{1}{2} u^{-3/2} du. \quad (50)$$

As σ varies from 0 to ∞ , u varies from ∞ to 0. Therefore,

$$\begin{aligned} &\int_0^\infty \sigma^{-(n-1)} \exp\left(-\frac{Q_\gamma}{2\sigma^2}\right) d\sigma \\ &= \frac{1}{2} \int_0^\infty u^{(n-1)/2} u^{-3/2} \exp\left(-\frac{Q_\gamma}{2} u\right) du \\ &= \frac{1}{2} \int_0^\infty u^{(n-4)/2} \exp\left(-\frac{Q_\gamma}{2} u\right) du. \end{aligned} \quad (51)$$

Using the Gamma integral

$$\int_0^\infty u^{a-1} e^{-bu} du = \frac{\Gamma(a)}{b^a}, \quad a > 0, \quad b > 0, \quad (52)$$

with

$$a = \frac{n}{2} - 1, \quad b = \frac{Q_\gamma}{2}, \quad (53)$$

we obtain

$$\begin{aligned}
& \int_0^\infty \sigma^{-(n-1)} \exp\left(-\frac{Q_\gamma}{2\sigma^2}\right) d\sigma \\
&= \frac{1}{2} \Gamma\left(\frac{n}{2} - 1\right) \left(\frac{Q_\gamma}{2}\right)^{-(n/2-1)} \\
&\propto Q_\gamma^{-(n/2-1)}.
\end{aligned} \tag{54}$$

The omitted proportionality constant depends only on n , and not on γ . Substituting (54) into (48) gives

$$p(\gamma | y) \propto p(\gamma) |A_\gamma|^{-1/2} |X^T A_\gamma^{-1} X|^{-1/2} Q_\gamma^{-(n/2-1)}. \tag{55}$$

Finally, substituting the definition of Q_γ gives

$$p(\gamma | y) \propto \frac{p(\gamma) |A_\gamma|^{-1/2} |X^T A_\gamma^{-1} X|^{-1/2}}{\left\{y^T \left[A_\gamma^{-1} - A_\gamma^{-1} X (X^T A_\gamma^{-1} X)^{-1} X^T A_\gamma^{-1}\right] y\right\}^{n/2-1}}. \tag{56}$$

Remark 5.2. The parameter $\gamma = \tau/\sigma$ controls the amount of smoothing in the fitted function. Large values of γ allow the integrated Brownian motion component to vary more freely, producing rougher fitted curves, while small values of γ shrink the IBM component toward zero, resulting in estimates that are close to a straight line.

5.2 The Bayesian Perspective on Tuning Parameters

One attractive feature of the Bayesian approach is that the amount of smoothing is incorporated directly into the probabilistic model. In frequentist methods such as smoothing splines, ridge regression, trend filtering, or kernel smoothing, one typically introduces a tuning parameter λ that controls the complexity of the fitted function. The value of *lambda* is then chosen using an additional procedure such as cross-validation, generalized cross-validation, AIC, BIC, or some other model-selection criterion.

In the Bayesian formulation, there is no separate tuning step. Instead, the amount of smoothing is determined by parameters of the prior distribution. In the Integrated Brownian Motion model, the relevant quantity is $\gamma = \frac{\tau}{\sigma}$, where τ controls the size of the IBM component and σ is the noise standard deviation. Rather than fixing γ , we treat it as an unknown parameter and infer it from the data through its posterior distribution.

The posterior distribution of γ quantifies which levels of smoothness are most compatible with the observed data. Once this posterior has been computed, inference for the regression function f automatically incorporates uncertainty about the smoothing parameter.

Thus, from a Bayesian point of view, regularization is not imposed through an externally chosen tuning parameter. Instead, it arises naturally from the prior distribution, and the amount of regularization can itself be learned from the data.

In practice, many implementations use an empirical Bayes approach, where γ is estimated by maximizing the marginal likelihood. A fully Bayesian analysis goes one step further by averaging over the posterior distribution of γ thereby propagating uncertainty about the appropriate level of smoothness into the final inference.

This illustrates a recurring theme in Bayesian statistics: quantities that appear as tuning parameters in frequentist procedures often become ordinary parameters in a probabilistic model and can be estimated as part of the inferential process.

Show GP Regression with IBM prior for the Wage dataset. One needs to invert A_γ , a $n \times n$ matrix. This is a potential bottleneck when n is large.

A Proof of Theorem 5.1

Proof. Recall that

$$\hat{f}(x) = (C(1, x)X^T + \tau^2 \mathbf{k}_I^T) \left[\Sigma^{-1} - \Sigma^{-1}X (C^{-1}I_2 + X^T \Sigma^{-1}X)^{-1} X^T \Sigma^{-1} \right] y.$$

Write

$$\hat{f}(x) = T_1 + T_2,$$

where

$$T_1 = C(1, x)X^T \left[\Sigma^{-1} - \Sigma^{-1}X (C^{-1}I_2 + X^T \Sigma^{-1}X)^{-1} X^T \Sigma^{-1} \right] y$$

and

$$T_2 = \tau^2 \mathbf{k}_I^T \left[\Sigma^{-1} - \Sigma^{-1}X (C^{-1}I_2 + X^T \Sigma^{-1}X)^{-1} X^T \Sigma^{-1} \right] y.$$

First consider T_2 . Since

$$(C^{-1}I_2 + X^T \Sigma^{-1}X)^{-1} \longrightarrow (X^T \Sigma^{-1}X)^{-1},$$

we get

$$T_2 \longrightarrow \tau^2 \mathbf{k}_I^T \left[\Sigma^{-1} - \Sigma^{-1}X (X^T \Sigma^{-1}X)^{-1} X^T \Sigma^{-1} \right] y.$$

Using

$$\Sigma = \sigma^2 A_\gamma, \quad \tau = \gamma \sigma,$$

this becomes

$$T_2 \longrightarrow \gamma^2 \mathbf{k}_I^T \left[A_\gamma^{-1} - A_\gamma^{-1}X (X^T A_\gamma^{-1}X)^{-1} X^T A_\gamma^{-1} \right] y.$$

Now consider T_1 . Let

$$B = X^T \Sigma^{-1}X.$$

Then

$$\begin{aligned} T_1 &= C(1, x) \left[X^T \Sigma^{-1} - B (C^{-1}I_2 + B)^{-1} X^T \Sigma^{-1} \right] y \\ &= C(1, x) \left[I_2 - B (C^{-1}I_2 + B)^{-1} \right] X^T \Sigma^{-1} y. \end{aligned}$$

Since B commutes with $C^{-1}I_2 + B$,

$$B(C^{-1}I_2 + B)^{-1} = (C^{-1}I_2 + B)^{-1}B.$$

Also,

$$(C^{-1}I_2 + B)^{-1}B = I_2 - C^{-1}(C^{-1}I_2 + B)^{-1}.$$

Therefore

$$I_2 - B(C^{-1}I_2 + B)^{-1} = C^{-1}(C^{-1}I_2 + B)^{-1}.$$

Hence

$$T_1 = (1, x)(C^{-1}I_2 + B)^{-1} X^T \Sigma^{-1} y.$$

Letting $C \rightarrow \infty$, we obtain

$$T_1 \longrightarrow (1, x) (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y.$$

Finally, since $\Sigma = \sigma^2 A_\gamma$, the factors of σ^2 cancel, giving

$$T_1 \longrightarrow (1, x) (X^T A_\gamma^{-1} X)^{-1} X^T A_\gamma^{-1} y.$$

Combining the limits of T_1 and T_2 proves the theorem. $\square$

B Mathematical Theory of Splines

The previous slides motivate splines as a flexible way of fitting nonlinear relationships while preserving much of the simplicity and interpretability of linear regression. We now study splines more systematically.

B.1 Regression Splines

Regression splines and smoothing splines are motivated from a different perspective than kernel methods and local polynomial regression. Instead of starting from local averages, we build the estimator globally from a collection of basis functions.

This is often called the *synthesis* viewpoint: we construct the function by combining simple building blocks, or atoms. In the present case, the atoms are spline basis functions.

B.1.1 Splines

For simplicity, we first consider the univariate case.

Definition B.1 (Spline). Let

$$t_1 < t_2 < \dots < t_r$$

be fixed knots. A spline of degree k with knots $t_1, \dots, t_r$ is a function f satisfying:

1. f is a polynomial of degree k on each of the intervals

$$(-\infty, t_1], \quad [t_1, t_2], \quad \dots, \quad [t_r, \infty);$$

2. for every $\ell = 0, \dots, k-1$, the derivative

$$D^\ell f$$

is continuous at each knot $t_1, \dots, t_r$.

Thus a spline is a piecewise polynomial with smoothness constraints at the knots. It is much smoother than an arbitrary piecewise polynomial, and this smoothness helps reduce the variance of the fitted estimator. A cubic spline is a spline of degree 3. It is piecewise cubic, and has continuous first and second derivatives at the knots.

Remark B.2. A useful piece of statistical folklore says that a cubic spline is so smooth that one often cannot detect the knot locations by eye.

B.1.2 Spline Bases

How do we parametrize the space of degree- k splines with knots $t_1, \dots, t_r$?

One natural parametrization is the *truncated power basis*. Define

$$x_+ = \max\{x, 0\}.$$

The truncated power basis functions are

$$g_j(x) = \frac{x^{j-1}}{(j-1)!}, \quad j = 1, \dots, k+1,$$

and

$$g_{k+1+j}(x) = \frac{(x - t_j)_+^k}{k!}, \quad j = 1, \dots, r.$$

Therefore any degree- k spline with knots $t_1, \dots, t_r$ can be written as

$$f(x) = \sum_{j=1}^{r+k+1} \beta_j g_j(x).$$

Equivalently,

$$f(x) = \beta_0 + \beta_1 x + \dots + \beta_k x^k + \sum_{j=1}^r \gamma_j (x - t_j)_+^k.$$

The dimension of the space of degree- k splines with r knots is therefore

$$r + k + 1.$$

For example, a cubic spline with knots $t_1, \dots, t_r$ has basis

$$1, x, x^2, x^3, (x - t_1)_+^3, \dots, (x - t_r)_+^3.$$

The truncated power basis is mathematically simple, but it can be numerically poorly conditioned. In practice, spline software typically uses the *B-spline basis*. B-splines have local support: for a degree- k spline, each B-spline basis function is supported only on a small number of neighboring knots. This makes computations faster and more stable.

B.1.3 Regression Splines

A first idea is to choose knots in advance and then perform least squares regression on a spline basis.

Let

$$g_1, \dots, g_{r+k+1}$$

be a basis for the space of degree- k splines with knots $t_1, \dots, t_r$. We model

$$f(x) = \sum_{j=1}^{r+k+1} \beta_j g_j(x).$$

Given observations

$$(x_1, y_1), \dots, (x_n, y_n),$$

let

$$Y = (y_1, \dots, y_n)^T$$

and define the basis matrix $G \in \mathbb{R}^{n \times (r+k+1)}$ by

$$G_{ij} = g_j(x_i).$$

Then the regression spline estimator is obtained by solving

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n \left(y_i - \sum_{j=1}^{r+k+1} \beta_j g_j(x_i) \right)^2.$$

Equivalently,

$$\hat{\beta} = \arg \min_{\beta} \|Y - G\beta\|_2^2.$$

Assuming $G^T G$ is invertible,

$$\hat{\beta} = (G^T G)^{-1} G^T Y.$$

The fitted function is

$$\hat{f}(x) = \sum_{j=1}^{r+k+1} \hat{\beta}_j g_j(x).$$

Writing

$$g(x) = (g_1(x), \dots, g_{r+k+1}(x))^T,$$

we get

$$\hat{f}(x) = g(x)^T (G^T G)^{-1} G^T Y.$$

Thus

$$\hat{f}(x) = w(x)^T Y$$

for

$$w(x) = G(G^T G)^{-1} g(x),$$

up to the convention of column versus row vectors.

Therefore regression splines are *linear smoothers*: the fitted value at any point is a linear combination of the observed responses.

Regression splines can work well if the knots are well chosen. The main practical difficulty is knot selection. Too few knots may underfit, while too many knots may overfit.

B.2 Smoothing Splines

Before defining smoothing splines, we introduce natural splines.

B.2.1 Natural Splines

Spline estimates can sometimes behave erratically near the boundary of the input domain. This boundary instability becomes more severe as the polynomial degree increases.

Natural splines reduce this boundary behavior by forcing the spline to have lower degree outside the boundary knots.

Definition B.3 (Natural spline). Let k be odd. A natural spline of degree k with knots

$$t_1 < \cdots < t_r$$

is a piecewise polynomial satisfying:

1. f is a polynomial of degree k on each interior interval

$$[t_1, t_2], \dots, [t_{r-1}, t_r];$$

2. f is a polynomial of degree

$$\frac{k-1}{2}$$

on the exterior intervals

$$(-\infty, t_1] \quad \text{and} \quad [t_r, \infty);$$

3. $D^\ell f$ is continuous at each knot for every

$$\ell = 0, \dots, k-1.$$

The most important case is $k = 3$. A natural cubic spline is cubic between knots and linear outside the boundary knots.

Let us count the dimension of the space of natural splines of degree k with knots $t_1, \dots, t_r$.

There are $r-1$ interior intervals, each requiring $k+1$ polynomial coefficients. This contributes

$$(k+1)(r-1)$$

parameters.

There are two exterior intervals, and on each exterior interval the degree is $(k-1)/2$. This contributes

$$2 \left(\frac{k-1}{2} + 1 \right)$$

parameters.

At each of the r knots, we impose continuity of derivatives of orders $0, \dots, k-1$, giving k constraints per knot. This subtracts

$$kr$$

constraints.

Thus the dimension is

$$(k+1)(r-1) + 2 \left(\frac{k-1}{2} + 1 \right) - kr = r.$$

This is a remarkable fact: the dimension is independent of k .

For cubic natural splines, this means that with r knots, the spline space has dimension r .

This also means that if we set r constraints $f(x_1) = y_1, \dots, f(x_r) = y_r$ then there is a unique natural spline interpolant of any degree k .

B.2.2 Smooth Away

Regression splines require choosing the knot locations. Smoothing splines avoid this problem by placing knots at all observed input points and controlling complexity through regularization.

Assume that

$$x_1, \dots, x_n \in [a, b].$$

For an odd integer $k \geq 1$, write

$$m = \frac{k+1}{2}.$$

The smoothing spline estimator is defined as the solution to

$$\hat{f} = \arg \min_f \left\{ \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_a^b (D^m f(x))^2 dx \right\}.$$

This is an infinite-dimensional optimization problem over functions for which the criterion is finite.

The first term is the residual sum of squares. The second term is a roughness penalty. It penalizes functions whose $(m-1)$ st derivative is too wiggly. The tuning parameter $\lambda \geq 0$ controls the tradeoff between fidelity to the data and smoothness.

The most common case is the cubic smoothing spline. Here $k = 3$ and $m = 2$, so the estimator solves

$$\hat{f} = \arg \min_f \left\{ \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_a^b (f''(x))^2 dx \right\}.$$

The penalty

$$\int_a^b (f''(x))^2 dx$$

measures curvature. A linear function has zero penalty, while a highly wiggly function has a large penalty.

When $\lambda = 0$, the estimator tries to interpolate the data. When λ is very large, the penalty dominates and the solution becomes very smooth, approaching a linear function in the cubic case.

B.2.3 Representer Theorem

At first glance, the smoothing spline problem looks difficult because it is an optimization problem over an infinite-dimensional function space. The key result is that the solution actually lies in a finite-dimensional spline space.

Theorem B.4 (Representer theorem for cubic smoothing splines). *The unique minimizer of*

$$\min_f \left\{ \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_a^b (f''(x))^2 dx \right\}$$

is a natural cubic spline with knots at the observed inputs

$$x_1, \dots, x_n.$$

Proof. We prove the main idea.

Let g be any function on $[a, b]$ with two derivatives and square integrable second derivative. We will show that there exists a natural cubic spline f with knots at $x_1, \dots, x_n$ such that

$$f(x_i) = g(x_i), \quad i = 1, \dots, n,$$

and

$$\int_a^b \{f''(x)\}^2 dx \leq \int_a^b \{g''(x)\}^2 dx.$$

Since the natural cubic spline space with knots at $x_1, \dots, x_n$ is n -dimensional, there exists a natural cubic spline f satisfying

$$f(x_i) = g(x_i), \quad i = 1, \dots, n.$$

Define

$$h = g - f.$$

Then

$$h(x_i) = 0, \quad i = 1, \dots, n.$$

We claim that

$$\int_a^b f''(x)h''(x) dx = 0.$$

By integration by parts,

$$\int_a^b f''(x)h''(x) dx = f''(x)h'(x) \Big|_a^b - \int_a^b f'''(x)h'(x) dx.$$

Because f is a natural cubic spline, it is linear outside the boundary knots x_1 and x_n . Hence

$$f''(a) = f''(b) = 0,$$

and

$$f'''(x) = 0$$

for $x \leq x_1$ and $x \geq x_n$. Therefore

$$\int_a^b f''(x)h''(x) dx = - \int_{x_1}^{x_n} f'''(x)h'(x) dx.$$

Now split the integral into the intervals (x_i, x_{i+1}) :

$$- \int_{x_1}^{x_n} f'''(x)h'(x) dx = - \sum_{i=1}^{n-1} \int_{x_i}^{x_{i+1}} f'''(x)h'(x) dx.$$

On each interval (x_i, x_{i+1}) , f is cubic, so f''' is constant. Thus

$$- \sum_{i=1}^{n-1} \int_{x_i}^{x_{i+1}} f'''(x)h'(x) dx = - \sum_{i=1}^{n-1} f'''(x_i^+) \{h(x_{i+1}) - h(x_i)\}.$$

Since $h(x_i) = 0$ for all i , every term in the sum is zero. Therefore

$$\int_a^b f''(x)h''(x) dx = 0.$$

Now use

$$g = f + h.$$

Then

$$g'' = f'' + h''.$$

Hence

$$\int_a^b \{g''(x)\}^2 dx = \int_a^b \{f''(x) + h''(x)\}^2 dx.$$

Expanding the square gives

$$\int_a^b \{g''(x)\}^2 dx = \int_a^b \{f''(x)\}^2 dx + \int_a^b \{h''(x)\}^2 dx + 2 \int_a^b f''(x)h''(x) dx.$$

The cross term is zero, so

$$\int_a^b \{g''(x)\}^2 dx = \int_a^b \{f''(x)\}^2 dx + \int_a^b \{h''(x)\}^2 dx.$$

Therefore

$$\int_a^b \{f''(x)\}^2 dx \leq \int_a^b \{g''(x)\}^2 dx.$$

Thus any candidate function g can be replaced by a natural cubic spline f that agrees with g at the data points and has no larger roughness penalty.

The data-fitting term depends only on the values

$$f(x_1), \dots, f(x_n).$$

Therefore the minimizer of the smoothing spline objective must be a natural cubic spline with knots at the observed inputs. This proves the representer theorem. $\square$

To summarize,

- Splines are smooth piecewise polynomials.
- A degree- k spline with r knots has dimension $r + k + 1$.
- Regression splines fit a spline basis by least squares.
- Natural splines improve boundary behavior by imposing linear tails in the cubic case.
- Smoothing splines avoid knot selection by putting knots at every data point and controlling complexity through a roughness penalty.
- The representer theorem shows that an infinite-dimensional optimization problem has a finite-dimensional natural spline solution.
- One can do cross validation to select the value of λ in smoothing splines.

Questions:

- We have defined natural splines in Definition [B.3](#). Is it obvious that this defines a linear space of functions?

- What is a set of basis functions for natural splines?

Answer. Yes, the definition does give a linear space. If f and g are natural splines of degree k with the same knots $t_1 < \dots < t_r$, then for any constants a, b , the function

$$af + bg$$

is again a polynomial of degree at most k on each interior interval, is a polynomial of degree at most $(k-1)/2$ on the two exterior intervals, and has continuous derivatives up to order $k-1$ at the knots. Therefore the class is closed under linear combinations, and hence is a linear space.

For cubic natural splines, the most important case is $k=3$. In this case the function is cubic between knots and linear outside the boundary knots. If the knots are

$$t_1 < \dots < t_r,$$

then the natural cubic spline space has dimension r . One convenient basis is

$$1, \quad x, \quad d_1(x), \quad \dots, \quad d_{r-2}(x),$$

where

$$d_j(x) = \frac{(x - t_j)_+^3 - (x - t_r)_+^3}{t_r - t_j} - \frac{(x - t_{r-1})_+^3 - (x - t_r)_+^3}{t_r - t_{r-1}}, \quad j = 1, \dots, r-2.$$

Thus every natural cubic spline can be written as

$$f(x) = \beta_0 + \beta_1 x + \sum_{j=1}^{r-2} \theta_j d_j(x).$$

This gives $2 + (r-2) = r$ basis functions, agreeing with the dimension count for natural cubic splines.

References

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